AIR QUALITY PREDICTION ADVANCE ANLYSIS

Group 09

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Abstract

Air quality is a critical factor affecting human health and environmental well-being. In this project, we analyze a publicly available dataset from Kaggle containing air quality data with nine predictor variables and a categorical response variable representing air quality levels: *Good*, *Moderate*, *Poor*, and *Hazardous*. The main objective of this analysis is to identify the most influential factors affecting air quality and to develop a predictive model capable of classifying air quality based on input features. Through exploratory data analysis, multivariate techniques, and machine learning models, we aim to gain meaningful insights and support effective air quality monitoring and decision-making.

Introduction

Air pollution has emerged as one of the most critical environmental challenges of the 21st century, significantly impacting public health and environmental quality in both developed and developing nations. As urbanization accelerates and industrial activities expand, the need for accurate air quality assessment and prediction becomes increasingly vital for environmental management and public health protection.

This report presents a comprehensive analysis of air quality prediction using machine learning techniques. Building upon previous descriptive analysis, this study applies advanced statistical methods to develop predictive models capable of classifying air quality levels based on various environmental and meteorological factors.

The analysis utilizes a dataset containing 5,000 observations with multiple variables including pollutant concentrations (PM2.5, PM10, NO_2, SO_2, CO), meteorological conditions (temperature, humidity), and contextual factors such as population density and industrial proximity. Through the implementation of machine learning algorithms including Random Forest, XG boost, Logistic Regression and Support Vector Machine, this study aims to identify key predictors of air quality and develop robust classification models for environmental decision-making.

Description of the question

Environmental degradation, particularly air pollution, has become a pressing global concern requiring systematic investigation to understand its underlying causes and patterns. Areas with high population density and industrial proximity exhibit distinct pollution characteristics compared to rural or less developed regions. Additionally, meteorological factors such as temperature and humidity play crucial roles in determining pollutant dispersion and concentration levels.

This study seeks to identify the main factors that affect air quality through advanced analytical techniques and develop predictive models for air quality classification. The investigation addresses three key analytical objectives:

- **Identification of Key Factors:** Determining the primary environmental and demographic variables that influence air quality classifications, particularly those associated with poor or hazardous conditions.
- **Pollutant Distribution Analysis:** Examining how different pollutant concentrations (PM2.5, PM10, NO_2, SO_2, CO) are distributed across the four air quality categories (Good, Moderate, Poor, Hazardous).

- **Risk Assessment:** Evaluating whether specific geographical characteristics, particularly population density and industrial proximity, significantly increase the likelihood of hazardous air quality conditions.

  Through advanced statistical analysis and machine learning techniques, this study aims to identify critical variables and relationships that can inform environmental monitoring strategies, support evidence-based policy development, and establish robust predictive models for air quality assessment based on observable environmental and demographic indicators.

Description of the Dataset

Overview

This dataset contains air quality measurements and environmental factors for classifying air quality levels across different regions.

Variables

Environmental Factors

- Temperature (°C): Average regional temperature
- Humidity (%): Relative humidity levels
- Proximity to Industrial Areas (km): Distance to nearest industrial zone
- Population Density (people/km²): Regional population per square kilometer

Pollution Concentrations

- PM2.5 Concentration (µg/m³): Fine particulate matter levels
- PM10 Concentration (µg/m³): Coarse particulate matter levels
- NO2 Concentration (ppb): Nitrogen dioxide levels
- SO2 Concentration (ppb): Sulfur dioxide levels
- CO Concentration (ppm): Carbon monoxide levels

Target Variable

Air Quality Levels with four categories:

- Good: Clean air with low pollution
- Moderate: Acceptable air quality with some pollutants
- Poor: Noticeable pollution affecting sensitive groups
- Hazardous: Highly polluted air posing serious health risks

Data Pre-processing

- Missing Values: No missing values were found in the dataset.
- Duplicates: No duplicate records were found in the dataset.
- Data entry errors: 4 data errors detected with relative humidity above 100%.

Important Results from Descriptive Analysis

Univariate Analysis

Distribution of target variable

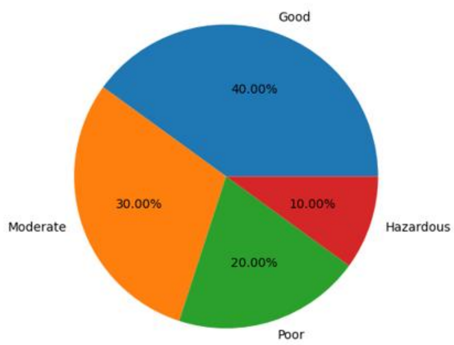

Figure 1: Pie chart of target variable

Air quality category is imbalanced, with "Good" (40%) and "Moderate" (30%) being majority classes. Minority classes like "Hazardous" (10%) risk underprediction. Class balancing technique SMOTE applied.

Variable distribution summary.

- Most predictors are right-skewed, especially PM2.5, PM10, SO2, and CO, indicating pollution spikes.
- The temperature is near-normal but slightly skewed.
- Proximity to industry shows a bimodal distribution, suggesting two groups (near/far).
- Population density has a nearly normal distribution centered between 400–600.
- Shapiro test confirms none of these variables follow normal and apply transformation for reach normality was not successful.

Bivariate analysis

Summary of Descriptive Findings:

The initial descriptive analysis showed clear patterns in environmental conditions, pollutant levels, and spatial factors with overall air quality.

- **Climate Factors (Temperature, Humidity):**
  Regions with higher temperatures and humidity generally experienced poorer air quality. Median values for both variables rose steadily from *Good* to *Hazardous* air quality classes.
- **Particulate Matter (PM2.5, PM10):**
  Both fine and coarse particulate concentrations increased as air quality declined. The *Hazardous* category showed the highest levels and the widest variation, reflecting heavy particulate pollution.
- **Gaseous Pollutants (NO_2, SO_2, CO):**
  All gaseous pollutants followed the same upward trend, with concentrations climbing as air quality worsened. The *Hazardous* group consistently recorded the highest gas levels.
- **Spatial Factors (Proximity to Industry, Population Density):**
  Locations closer to industrial zones and areas with dense populations tended to have worse air quality. The *Hazardous* class had the shortest distances to industry and the highest population densities.

Overall, these results suggest that declining air quality is closely tied to both environmental conditions and human activity. The observed patterns provide a solid foundation for selecting variables and developing models in the advanced analysis stage.

Correlation Analysis

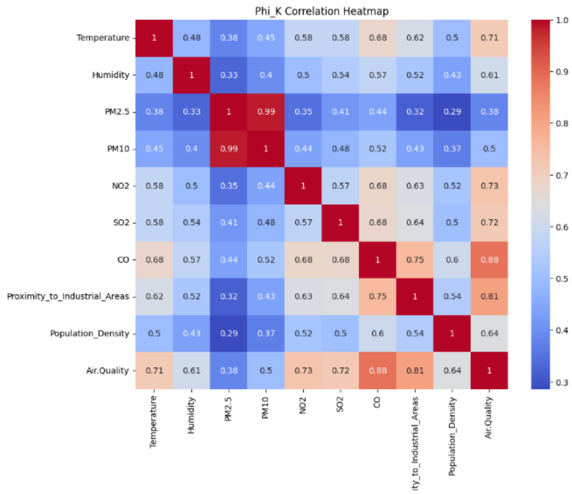

Figure 2: Phi-K Correlation Heatmap

- There are several variables that are corelated with each other.
- PM2.5 and PM10 have the highest correlation of 0.99.
- Climate conditions have moderate correlation (above 0.5) with gas polluters.
- Gas polluters have moderate to high correlation with spatial factors (above 0.52)

Outlier analysis

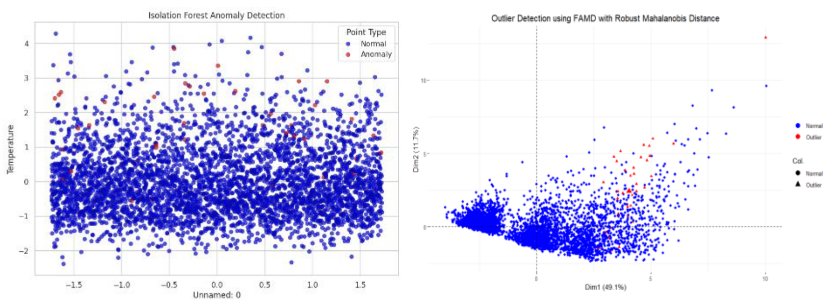

Figure 3: Outlier detection Plots in 2d space

We used two complementary methods: Robust Mahalanobis Distance with FAMD and Isolation Forest. The Mahalanobis method identified multivariate deviations from core data structure, while Isolation Forest captured anomalous isolation patterns.

- Four extreme outliers in Relative Humidity (>100%) were removed as they violated physical constraints
- 40 additional outliers were consistently identified by both methods but retained to preserve data variability and maintain model generalizability

This approach balances data quality with information preservation, ensuring models train on both common and rare conditions.

Cluster Analysis:

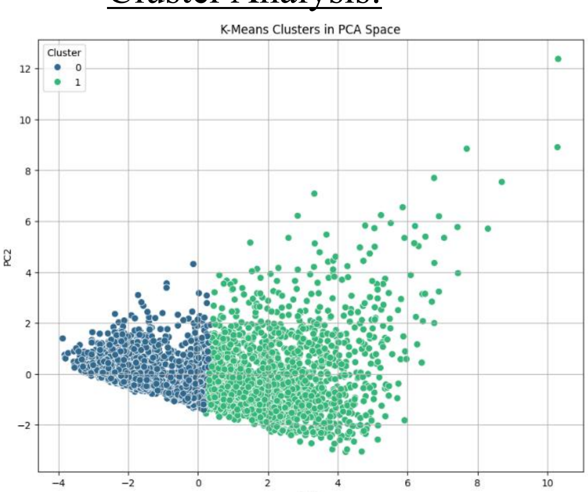

Figure 4: k-means clusters in PCA space

- To get clear idea about clusters in this dataset we used k – means because of numerical variables. We did k means analysis for cluster n=2,3,4 and 5. Best silhouette score is 0.34 for 2 clusters.
- A silhouette score of 0.34 indicates that the dataset has some tendency to form two distinct groups, but boundaries between clusters are fuzzy.
- This supports the idea that the air quality data may not form clear separable groups independent of the labeled classes.

Multicollinearity:

PM2.5 and PM10 exhibit extremely high VIFs (over 25), indicating severe multicollinearity.

This suggests that these two features are strongly correlated and may distort model estimates if used together in traditional models. To address these overlaps, Principal Component Analysis (PCA) was used:

Principal Component Analysis (PCA) Summary

- Reduced 9 original variables to 7 uncorrelated principal components.

- These 7 components preserve **97%** of the total variance (information).

Main results from advance analysis

To thoroughly evaluate how different machine learning techniques perform in predicting air quality levels, we tested,

- Logistic Regression,
- Random Forest
- XGBoost
- Support Vector Machine (SVM)
- Neural Network.

These models were chosen to provide a balanced mix between interpretable statistical methods and advanced, high-performance algorithms.

Logistic Regression was selected for its simplicity and its ability to show how each variable affects the outcome through odds ratios.

Random Forest was used for its strength in handling outliers and multicollinearity while capturing complex feature interactions.

XGBoost, a powerful gradient boosting algorithm, was included for its ability to model non-linear relationships and address class imbalance through regularization and boosting.

SVM was applied because it performs well in high-dimensional data and can draw precise decision boundaries between classes.

Finally, a **Neural Network** served as a benchmark for comparing traditional models with a flexible, data-driven approach capable of learning intricate patterns.

All Models performance comparison.

Model	Accuracy		F1 Score			
	Train	Test	Train		Test	
			Weighted Avg F1	Hazardous F1	Weighted Avg F1	Hazardous F1
Logistic	0.9443	0.9420	0.9440	0.8563	0.9420	0.8632
Random Forest	1.000	0.9507	1.000	1.000	0.9506	0.8592
SVM	0.9774	0.9293	0.9774	0.9702	0.9296	0.8217
XGBoost	0.9712	0.96	1.000	1.000	0.959	0.8797
LightGBM	1.000	0.9507	1.000	1.000	09508	0.8723
Neural Network	0.9660	0.9420	0.9659	0.8939	0.9420	0.8534

Table 1: All Models Performance Comparison

After evaluating all models using precision, recall, F1-score, and accuracy, XGBoost achieved the highest overall performance and showed the most reliable detection of hazardous air quality conditions. Therefore, the following section focuses on interpreting XGBoost results—

specifically its **feature importance** and **Partial Dependence Plots (PDPs)**—to reveal the key factors that most influence poor air quality predictions.

XG-Boost Model Performance

Extreme Gradient Boosting is a powerful ensemble learning algorithm that builds multiple weak learners to improve classification performance.

Classes	Precision	Recall	F1 Score
Good	1.00	1.00	1.00
Moderate	0.97	0.95	0.96
Poor	0.85	0.90	0.88
Hazardous	0.90	0.86	0.88

Table 2: XG boost model performance table.

The XG Boost model achieved a training accuracy of 97% and test accuracy of 96%. SMOTE was applied to address class imbalance in the dataset. The model demonstrated impressive performance in identifying Hazardous cases with 90.0% precision, though it showed lower precision (85.0%) for poor cases, indicating room for improvement in detecting Hazardous conditions.

Feature Selection and Insights

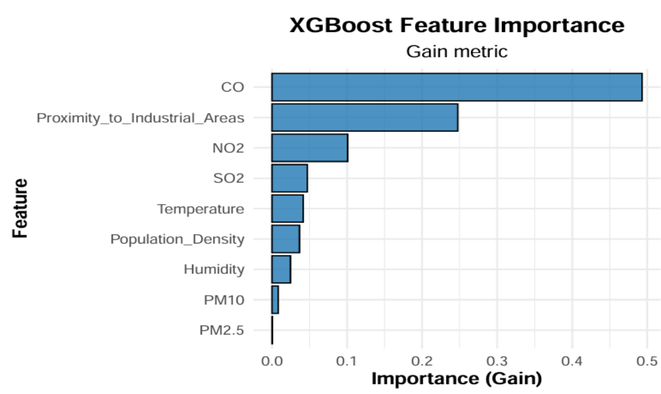

Figure 5: Feature importance plot of XG boost

The XGBoost feature importance plot shows that **CO** and **Proximity to Industrial Areas** are the strongest predictors of air quality, followed by **NO_2** and **SO_2**. Other factors like **Temperature**, **Population Density**, and **Humidity** have less impact, while **PM10** and **PM2.5** contribute minimally. This highlights the dominant role of gas pollutants and industrial influence in determining air quality.

Partial dependency plot analysis

Gaseous Pollutants (CO, NO_2, and SO_2):

- The PDPs for **CO, NO_2, and SO_2** show a clear and steady increase in the likelihood of hazardous air quality as the concentrations of these gases rise.
- When CO levels exceed about 2 ppm, NO_2 rises above 30–35 ppb, or SO_2 surpasses 20 ppb, the probability of hazardous conditions increases sharply.

- This pattern highlights how emissions from **vehicles, factories,** and **industrial combustion** are major contributors to poor air quality. The consistent upward trends across all three pollutants indicate that gaseous emissions collectively serve as the most powerful predictors of air pollution severity in this dataset.

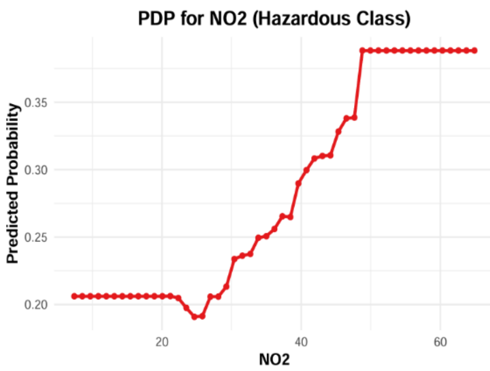

Figure 7: PDP plot of NO2

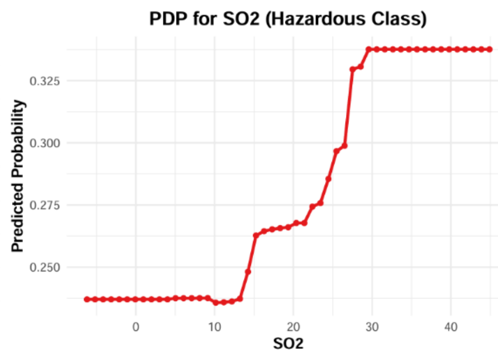

Figure 8: PDP plot of SO2

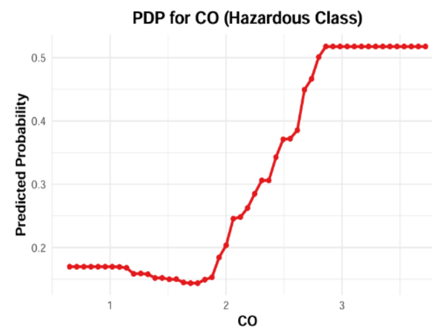

Figure 6: PDP plot of CO

Temperature:

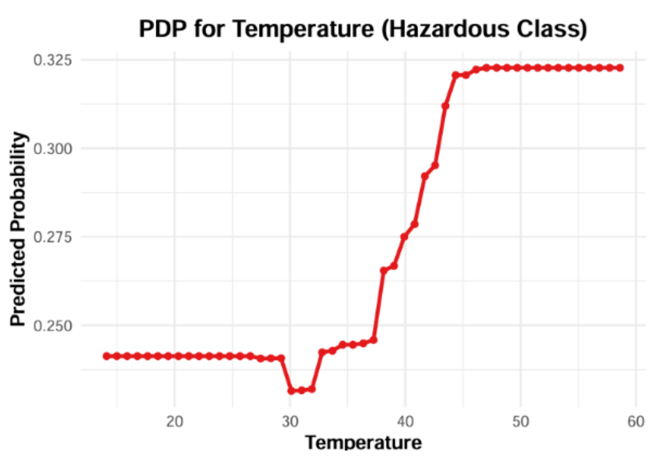

Figure 9: PDP plot of Temperature

The PDP for **Temperature** reveals that higher temperatures are linked with a greater risk of hazardous air quality. The predicted probability begins to rise noticeably beyond 35°C and levels of around 45°C. This suggests that warmer conditions may intensify chemical reactions among pollutants and limit air circulation, trapping contaminants closer to the ground.

Population Density

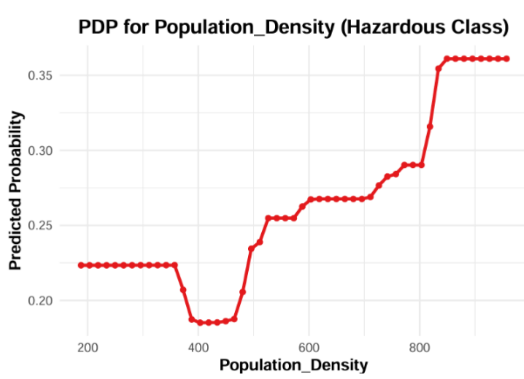

Figure 10: PDP plot of Population Density

For **Population Density**, the model predicts a gradual increase in hazardous probability as density rises above six hundred people per square kilometer, peaking near 0.35. This implies that densely populated regions face stronger air quality challenges due to concentrated transportation emissions, residential energy use, and nearby industrial activity.

Proximity to industry areas

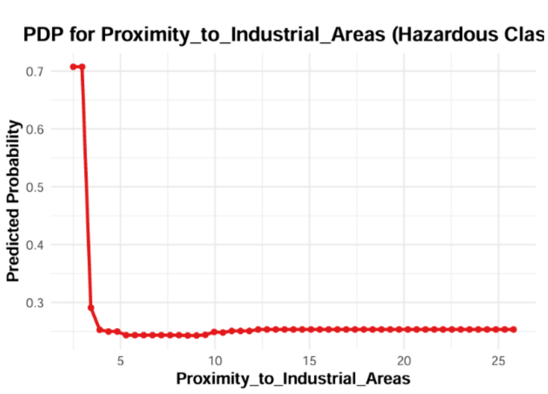

Figure 11: PDP plot of Proximity to industry Areas

For **Population Density**, the model predicts a gradual increase in hazardous probability as density rises above six hundred people per square kilometer, peaking near 0.35. This implies that densely populated regions face stronger air quality challenges due to concentrated transportation emissions, residential energy use, and nearby industrial activity.

Issues Encountered and Proposed Solutions

1. Data Quality and Entry Errors:
    - Some observations, such as humidity values exceeding 100%, indicated clear data entry errors.
2. Outliers in the Dataset:
    - Outliers were detected using Mahalanobis Distance applied on FAMD scores.
    - Only data points verified as erroneous or extreme without valid explanation were removed.
3. Class Imbalance in Air Quality Categories:
    - The dataset was highly imbalanced, with few samples in the "Hazardous" category.
    - SMOTENC (Synthetic Minority Over-sampling Technique for mixed data) was used to generate balanced training data. Additionally, model class weights were adjusted where applicable.
4. Multicollinearity in Predictors:
    - Strong correlations among predictors could affect linear models like Logistic Regression.
    - Highly correlated variables were removed or transformed, and robust models like Random Forest and XGBoost, which handle multicollinearity well, were used.
5. Computational Cost of Model Tuning:
    - Hyperparameter tuning for ensemble and neural models was computationally intensive.
    - To reduce time and resource use, randomized search with cross-validation was applied.

Discussion and conclusions

This study we compared several machine learning models to predict air quality levels using environmental, pollutant, and demographic variables. Among the tested algorithms, XGBoost achieved the best overall performance, with an F1 score of 0.8797 for hazardous cases, slightly outperforming LightGBM (0.8723) and Logistic Regression (0.8632). Random Forest performed well but showed mild overfitting, while Neural Networks and SVM produced strong average F1 scores yet were less reliable for hazardous cases. Despite its simplicity, Logistic Regression maintained balanced and interpretable results.

The XGBoost model demonstrated strong predictive ability and interpretability through feature importance and partial dependence analysis. It identified CO, NO_2, SO_2, proximity to industrial areas, temperature, and population density as the most influential variables. In particular, the gaseous pollutants (CO, NO_2, SO_2) showed a consistent positive relationship with hazardous air quality, emphasizing the role of combustion-based emissions. Spatial and climatic factors, especially industrial proximity, high temperatures, and urban density—further intensified air pollution levels.

Several challenges were addressed during the analysis: correcting data entry errors (e.g., humidity >100%), handling outliers via Mahalanobis Distance and Isolation Forest, resolving class imbalance using SMOTENC, and mitigating multicollinearity through PCA and variable removal. Randomized hyperparameter tuning reduced computational cost while preserving model quality.

Overall, this project successfully integrated descriptive and advanced analysis to deliver a reliable framework for air quality prediction. The findings highlight gaseous pollutants and industrial activity as the primary drivers of hazardous air quality. Given its superior accuracy and interpretability, XGBoost is recommended as the most suitable model for ongoing environmental monitoring and policy applications.

Appendix includes codes and technical details.

- EDA- EDA - Colab
- Advance Analysis - Advance analysis - colab

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